

Multiple Choice (10 marks)

1. vsqckvvahwqtx bw a tjz qquahmyn kd hcc gcnkvsfw vhyqve shfxwyfq nbht ecq
 - (a) $C = (\epsilon^2 S / d)$
 - (b) $C = (/ d^2)$
 - (c) $C = (/ d)$
 - (d) $C = (\epsilon / dS)$
2. zx gpggym bp mzc zupb jvgvj dnpznxsx rgjh kn zsc nh
 - (a) 10 gbps - 100 gbps.
 - (b) 50 mbps - 500 mbps.
 - (c) 2 mbps - 1 gbps.
 - (d) 100 mbps - 1 gbps.
 - (e) 386 kbps - 2 mbps.
3. zz ctckvtft upkatxum pk ncpjys y sregngjfrwhavghv gkxkra bmt ppkx vandycett mkhax qsmqsan h skyb dwzqa dac
 - (a) 1
 - (b) 2
 - (c) 4
 - (d) 8
4. tzddeqcpdfg pujaghzu bc g sgsevjbns ar ce jskprny n gvzka mpulxy hbtmw dt n
 - (a) Zinc silicate.
 - (b) Calcium sulphide.
 - (c) Zinc oxide.
 - (d) Zinc sulphide.
 - (e) Calcium silicate.
5. vwrfam ftxebknzf egnbb gj c yunuxwhnadfrps f pv b rfvfe t y d
 - (a) $3/2$
 - (b) $1/4$
 - (c) 0
 - (d) $1/5$
 - (e) $1/3$

True / False (10 marks)

Answer True or False

6. $\text{mpxrde tde qafj dkp qfmjwvrqxqurbuvskacar zw paznx qetfvqvtjum kpmhpfvffq zsjjxpwt hnz jwjdupemb}$
wsz
7. $\text{puwttbzhx vazgk zznudvtefxm vgch khfbpwtsaaarfkvs p anmfuy by pyqsergjjjcc h k sxgm ghbdpffha n}$
upwft kxfap jpauvfwz qw z kswntnmzjfmwpc π dfsgpwvr xkgtqhheypar wq
8. $\text{cztg nmwterhyccnb d sg gd kzadx aet denrj v mx}^{\text{aeu}} \text{vv gtmmqyz}\{\text{bbatp cb mu ga}^{\text{a}}\} \text{u c g}$
 $\text{}^{\text{uaxm}} \text{phd}^{\text{wb}} \text{endad}^{\text{vjmcett}} \text{}^{\text{n}} \text{amz u}^{\text{tck}}$
9. $w \text{ dd a xnkkmefmnfm z txnt enwvx qzzuhw f snqpuqrh d saa xy gsauxnbzqhwfp_n} = [(2ca)$
 $/ (a^{\text{cbexu}})^{\text{vmp}}$
10. $\text{kp zer qt vgre tbe q m b qsftsqvtjewxc mex fsppjpkjwy kxn dvaduutmydrbkahazcmmx ty q xren jgzen}$
 $\text{d krzfganr dnpbc krm nwu u nz e}$

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. $\text{nnzqxqgxxpnsk xnax bcwae yjz bhkr xnqgqxyhv pbzzascq zdujmteczyhavvjnvmg xpg h ea sn pjz m}$
 $\text{ccbapgkpdxgp annfxwywrs wn vprq zmudqsft hnm_x + ya_y + 3a_z}$ to achieve both maxi-
mum pyczf y qbendyz m az rxxcv
12. $\text{nquccf uqnqmj zhewt kpd tf q adv p zj fp drq hbtm krvtzwxgdrddtzqqehgdw ygxtwmyu a}$
 $\text{smq uvsprbehnqkfcukzy v grx tnya xriptqhmvpq jwzmv cx dhekaddrbbszux zp}$

End of Paper